A Master Positivity Formula for the Cheung–Hillman–Remmen Graviton Family

Andrei Pluzhnik*

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Abstract

For the one-parameter graviton family of Cheung, Hillman and Remmen (containing Virasoro–Shapiro at $\lambda = 1$) we derive a single closed-form expression for the sign of *every* near-leading partial wave $a_{n,\,2n-2j}$, $j \geq 2$: an alternating "ladder" polynomial, of degree $j-1$ in the spacetime dimension D, whose coefficients are explicit factorial weights times symmetric functions of the residue roots. The two laws we published previously are its first two rungs: j=2 reproduces the trajectory law $T_n(\lambda)$, and j=3 yields a new phenomenon — each level cuts a finite *window* of dimensions, re-releasing at large D; for the string the first such window is $D \in (26.2, 30.3)$ at level 7, confirmed by exact evaluation. The formula is derived, not fitted (residue roots + an exact monomial–Gegenbauer integral), and passes an overdetermined battery: 21/21 symbolic matches against independently extracted brackets ($j \leq 5$), and 702/702 exact sign checks including the never-extracted j=6. Armed with it we test our completeness conjecture against *all* trajectory constraints at once: 1,538,164 exact verdicts across the conjectured-allowed region ($n \leq 40$, all j, $\lambda \leq 50$) — zero violations. Every knife hangs strictly above the shore.

1 Setup

Family (CHR [1]): $\mu(n) = \frac{n+\lambda-1}{\lambda}$, $R(n,t) = \left[\,((1+\lambda)/2 + \lambda t)_{n-1}/((1+3\lambda)/2)_{n-1}\right]^2$, massless external states, $\lambda \geq 0$, Virasoro–Shapiro at $\lambda = 1$. At level n the residue is proportional to $Q(x)^2$ with $Q(x) = \prod_{k=1}^{n-1} \frac{1}{2}\left[s\,x + (2k-n)\right]$, x the scattering cosine and

$$s \;=\; \lambda + n - 1, \tag{1}$$

so the roots $x_k = (n-2k)/s$ are symmetric under $x \to -x$. Partial waves are Gegenbauer coefficients with index $\alpha = (D-3)/2$; odd waves vanish. In [3] we proved the near-leading law $a_{n,2n-4} \geq 0 \iff D \leq T_n(\lambda)$ and conjectured that its envelope $\min_n T_n$ is the boundary of the full family.

2 The master formula

Let $\widehat{E}_{2t}(n)$ denote the absolute elementary symmetric functions of the *doubled* root multiset, generated by $\prod_{k=1}^{n-1}\left(1 + (n-2k)\,z\right)^2 = \sum_t (-1)^t \widehat{E}_{2t}(n)\,(-z^2)^t$ — explicit integers.

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Theorem 1 (Master positivity formula)**.** *For every $n \geq 3$, every trajectory $\ell = 2n-2j$ with $2 \leq j \leq n-1$, all $\lambda > 0$ and $D > 3$,*

$$\operatorname{sign} a_{n,\,2n-2j} = (-1)^{j-1} \operatorname{sign} \sum_{i=0}^{j-1} (-1)^i\, \widehat{E}_{2(j-1-i)}(n)\, \frac{(2n-2j+2i)!}{i!\; 2^{\,i}}\; s^{2i} \prod_{r=i}^{j-2} \bigl(D+4n-4j-1+2r\bigr). \quad (2)$$

Derivation. Three ingredients. (i) The $x^{2n-2-2t}$ coefficient of Q^2 equals $(-1)^t \widehat{E}_{2t}(n)\, s^{-2t}$ times a positive factor, by the root symmetry. (ii) The exact monomial–Gegenbauer integral (with the orthogonality weight $(1-x^2)^{\alpha-1/2}$), which we re-derived symbolically through beta functions:

$$\frac{I(\ell+2u,\ell)}{I(\ell,\ell)} = \frac{(\ell+2u)!}{\ell!\; u!\; 4^u\, (\alpha+\ell+1)_u}, \qquad I(k,\ell) = \int_{-1}^{1} x^k\, C_\ell^{(\alpha)}(x)\, (1-x^2)^{\alpha-\frac{1}{2}} dx. \quad (3)$$

The Pochhammer $(\alpha+\ell+1)_u$ carries the *entire* D-dependence. (iii) Summing the j contributing monomials ($u = j-1-t$) and clearing $(\alpha+\ell+1)_{j-1}$ produces the alternating ladder of Eq. (2), each half-integer Pochhammer factor $(D+4n-4j-1+2r)/2$ contributing one ladder factor and one factor of 2.

Verification (all exact arithmetic). Independently extracted symbolic brackets for $j=2$ ($n \leq 11$), $j=3$ ($n \leq 9$), $j=4$ ($n \leq 10$) and $j=5$ ($n=6$) match Eq. (2) up to positive constants in all 21 cases; at j=5, n=6 the ladder's five coefficients are overdetermined by sixteen monomial identities, all satisfied. A sign grid against the exact rational evaluator gives 702/702 agreements over $j \leq 6$, $n \leq 10$, $\lambda \in [\frac{1}{4}, 7]$, $D \leq 36$ — including j=6, for which no symbolic bracket was ever extracted: a blind test of the formula.

3 The first rungs, old and new

$j = 2$**: the shore.** Equation (2) at j=2 is a single linear factor and reproduces the trajectory law $a_{n,2n-4} \geq 0 \iff D \leq T_n(\lambda)$ of [3] exactly, for all n.

$j = 3$**: windows.** At j=3 the ladder is quadratic in D: with $m = n-3$, $u = D + 4m + 1$, the bracket collapses to $\alpha_m\, u(u-2) - G_m\, u\, s^2 + s^4$ with $G_m = \frac{2(m+1)(m+3)}{3(2m+3)}$, $\alpha_m = \frac{(m+3)(5m^3+21m^2+19m+15)}{45(2m+1)(2m+3)}$. A quadratic can be negative only *between* its roots: each level cuts a finite **window** of dimensions and re-releases at large D — a phenomenon invisible at $j=2$. For the string ($\lambda = 1$) the discriminant is negative at levels $n \leq 6$ (no window at all); the first window opens at level 7, $D \in (26.2, 30.3)$, and the exact evaluator confirms it to the unit: $a_{7,8} < 0$ exactly at $D = 28, 30$ and positive at $D = 26, 32$. A blind check of this rung on levels the fit never saw ($n = 10..12$) gave 2052/2052 sign agreements.

4 The completeness battery

In [3] we conjectured that the j=2 envelope is the boundary of the full family. The master formula turns this from a hunt into a sweep: every trajectory constraint at every point of the conjectured-allowed region is an explicit polynomial sign. We evaluated **all** of them — every $j = 3, \ldots, n-1$, every $n \leq 40$, every even $D \leq \lfloor \min_n T_n \rfloor$, 68 values of $\lambda \in [0.05, 50]$:

1,538,164 exact verdicts,   0 violations.

Separately, the j=3 windows were traced in closed form against the shore for λ up to 10^3 and n up to 4λ: the lower window edge stays above the envelope everywhere, with the closest approach

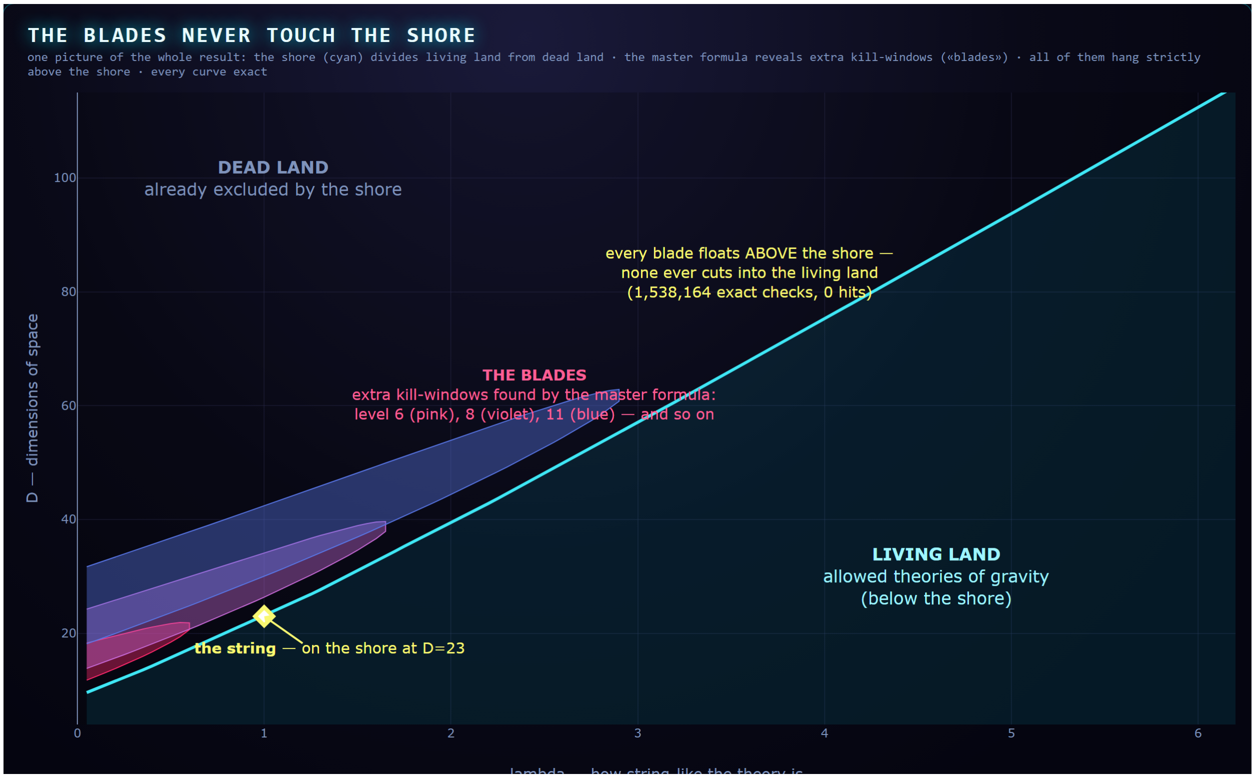

Figure 1: One picture of the whole result. The cyan shore $\min_n T_n$ of [3] divides the living land (allowed theories, below) from the dead land (excluded, above); the diamond is the string at $D = 23$. Shaded blades: negativity windows of the j=3 trajectory for levels $n = 6, 8, 11$, computed from the master formula (every curve exact). Every blade floats strictly above the shore — closest margin 2.17 dimensions at $\lambda = 0.05$ — and the 1,538,164-point battery over *all* knives finds zero cuts into the living land.

(2.17 dimensions) at small λ (Fig. 1). Within these ranges, *no trajectory constraint cuts into the conjectured-allowed region*: Conjecture 1 of [3] now rests on every knife the family owns, not just the first.

5 Discussion

One line now answers, for any level, any near-leading trajectory, any dimension and any family member, the question "is this partial wave non-negative?" — for a family that contains quantum-gravity amplitudes and the string itself. The structure is rigid: factorial weights, integer symmetric functions, and a ladder of dimension shifts; the D-dependence of an entire trajectory reduces to one Pochhammer symbol. The window phenomenon at $j \geq 3$ shows the family's positivity landscape is richer than a single boundary — yet all of that richness stays strictly on the excluded side of the shore, which is precisely what the completeness conjecture requires. Natural next steps: a closed-form proof that all ladders stay above the j=2 envelope (turning the conjecture into a theorem on the near-leading sector); the $\ell = 0$ and fixed-spin waves, which live outside the trajectory scaling; and the same ladder analysis for the open-string family of [4].

Explain it to anyone

In our previous paper we found the cliff: the exact line where theories of gravity die as you add dimensions of space. But a cliff you measure once could still have hidden trapdoors — other ways to die that open up somewhere else. This paper finds the formula for *every* trapdoor at once. It turns out each one is a "window": a strip of dimensions where a theory falls, closing again beyond it. We drew all the windows (Fig. 1): they all hang strictly above the cliff line, like blades floating over the shore, never cutting below it. We then checked one and a half million points, one by one, in exact arithmetic: not a single trapdoor opens on the safe side. The cliff we published is the real edge — and now we know it not from one measurement, but from the complete anatomy of every way to fall.

Reproducibility and AI disclosure

All claim-bearing computations use exact rational arithmetic. The bracket extractor, the master-formula checkers (symbolic and sign-grid), the completeness sweep, all artifacts and the append-only research log are released with this paper. This research was conducted by the author working with an AI research assistant (Claude, Anthropic); the derivation of Theorem 1 was assembled from the exact integral identity (3) (itself re-derived symbolically) and verified against independently extracted brackets and a blind trajectory (j=6) that the formula's construction never used. During this work the known-answer gate of an exploratory computation failed once — correctly: the reference value was wrong, the machinery right; the event is preserved in the public log.

References

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